Stat 4307 HW4

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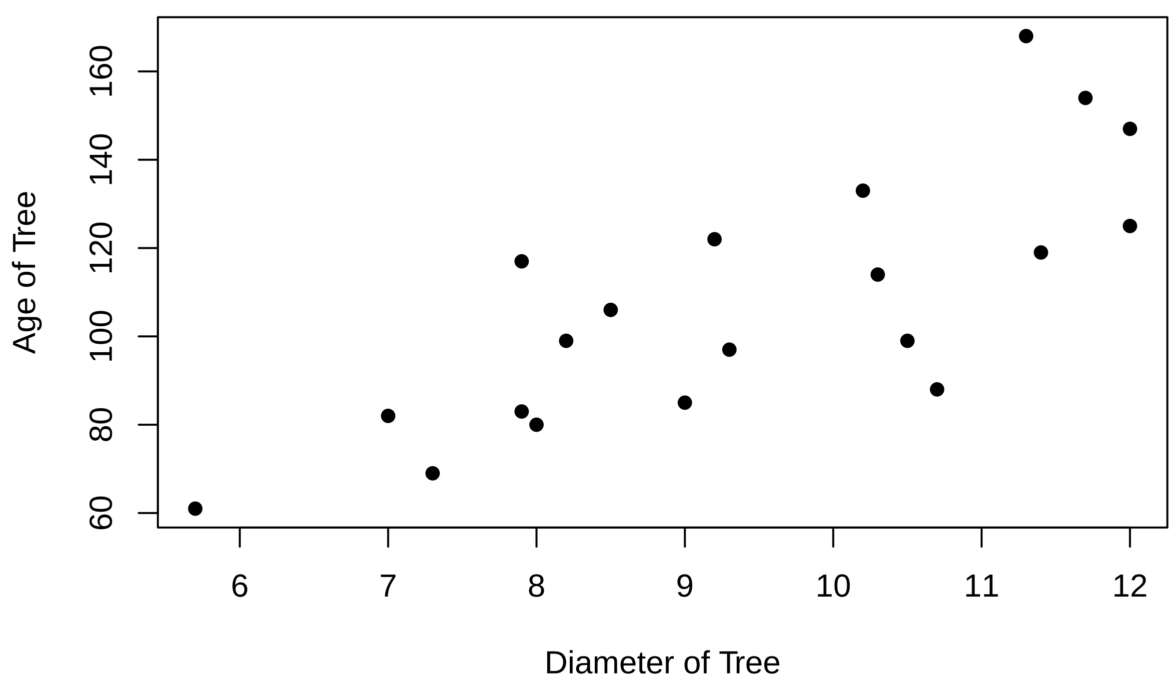
Chap 4

3: Foresters want to estimate the average of trees in a stand. Determining age is cumbersome. . . .

a. Draw a scatterplot of y vs x

```
x<-c(12,11.4,7.9,9.0,10.5,7.9,7.3,10.2,11.7,11.3,5.7,8.0,10.3,12.0,9.2,8.5,7.0,10.7,9.3,8.2)
y<-c(125,119,83,85,99,117,69,133,154,168,61,80,114,147,122,106,82,88,97,99)
plot(x,y, xlab="Diameter of Tree", ylab="Age of Tree", pch=16)
```




```
df<-as.data.frame(x=x,y=y)
```

b. Estimate the population mean age of the trees in the stand using ratio estimation and give and approximate standard error for your estimate.

Using the formulas for Ratio Estimation in a SRS
$t_y = \Sigma y_i$
$t_x = \Sigma x_i$
and the ratio is
B=$t_y/t_x = \bar{y}_u/\bar{x}_u$
$\widehat{B} = \bar{y}/\bar{x} = \widehat{t}_y/\widehat{t}_x$
$\widehat{t}_{yr} = \widehat{B}t_x$
$\widehat{\bar{y}}_r = \widehat{B}\bar{x}_U$
$e_i = y_i - \widehat{B}x_i$
$s_e^2 = 1/(n-1)\Sigma e_i^2$

```
ty<-sum(y)
tx<-sum(x)
b<-ty/tx
ybar<-mean(y)
xbar<-mean(x)
hatb<-ybar/xbar
thatyr<-hatb*(tx)
yhatbarr<-hatb*xbar
yhatbarr
```

```
## [1] 107.4
```

```
ybar
```

```
## [1] 107.4
```

```
thatyr2<-ybar*(tx/xbar)
thatyr
```

```
## [1] 2148
```

```
thatyr2
```

```
## [1] 2148
```

```
N<-1132
e<-c()
for(i in 1:length(y)){e[i]<-y[i]-(hatb*x[i])}
e
```

```
##  [1] -12.033493 -11.181818  -7.213716 -17.775120 -20.904306  26.786284
##  [7] -14.362041  16.521531  20.392344  38.960128  -4.090909 -11.355662
## [13]  -3.620415   9.966507  16.940989   8.934609   2.063796 -34.188198
## [19]  -9.200957   5.360447
```

```r
e2<-e^2
e2
```

```
##  [1]  144.804950  125.033058   52.037700  315.954877  436.990019  717.505005
##  [7]  206.268235  272.960990  415.847714 1517.891542   16.735537  128.951057
## [13]   13.107402   99.331265  286.997103   79.827242    4.259253 1168.832867
## [19]   84.657609   28.734387
```

```r
s2<-(1/(20-1))*sum(e^2)
s2
```

```
## [1] 321.933
```

```r
xt<-N*xbar
yt<-N*ybar
se<-1132*sqrt(1-(20/1132))*((yt)/(thatyr))*(s2/sqrt(20))
se
```

```
## [1] 4571330
```

Mean age of the trees using Ratio Estimation: 107.4 Standard Error of Estimate: 4571330

c. Repeat (b) using Regression

```r
lin.mod<-lm(y~x)
lin.mod
```

```
## 
## Call:
## lm(formula = y ~ x)
## 
## Coefficients:
## (Intercept)            x  
##      -7.808       12.250
```

```r
summary(lin.mod)
```

```
## 
## Call:
## lm(formula = y ~ x)
## 
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -35.263 -12.670  -2.689  11.230  37.387 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)    
## (Intercept)   -7.808     22.056  -0.354    0.727    
## x             12.250      2.304   5.316 4.71e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.37 on 18 degrees of freedom
## Multiple R-squared:  0.6109, Adjusted R-squared:  0.5893
## F-statistic: 28.26 on 1 and 18 DF,  p-value: 4.705e-05
```

```
xbar
```

```
## [1] 9.405
```

```
ybar
```

```
## [1] 107.4
```

```
yhatreg<-(-7.808)+12.250*(xbar)
yhatreg
```

```
## [1] 107.4032
```

```
rsids<-residuals(lin.mod)
vary<-var(rsids)
vary
```

```
## [1] 319.6277
```

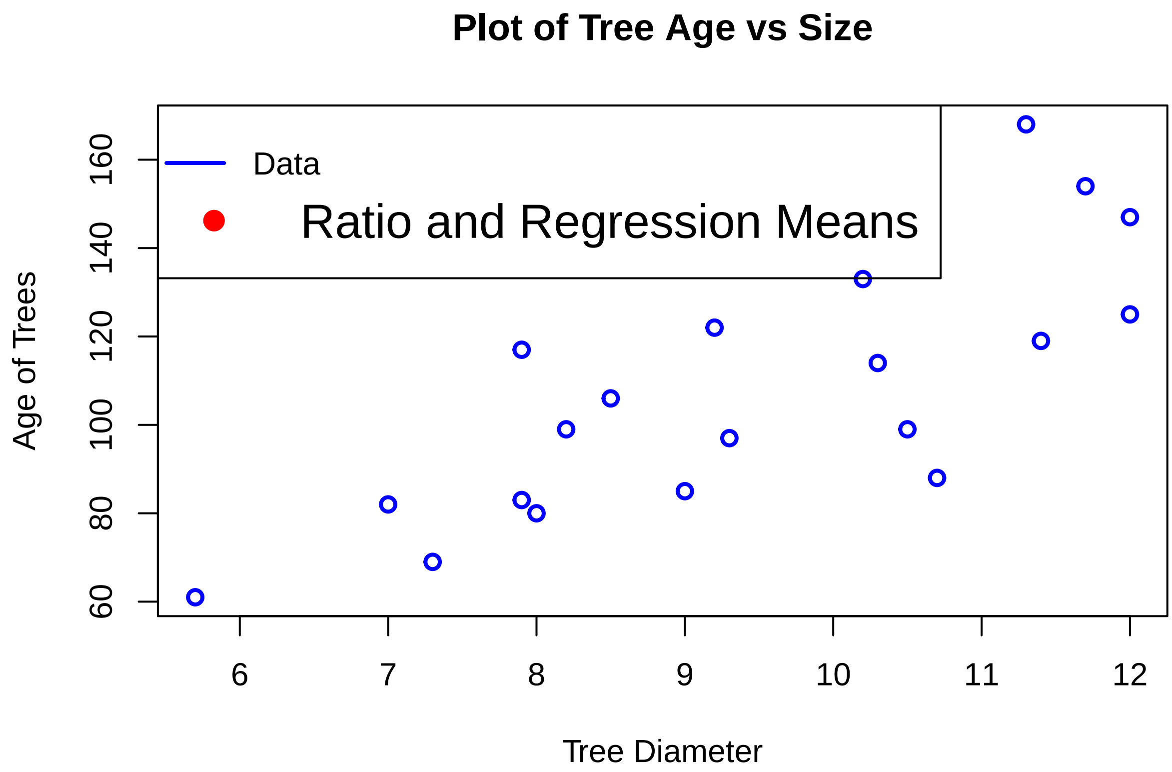
```
SE<-sqrt((1-(20/1132))*(vary)*(1-0.5893^2))
SE
```

```
## [1] 14.31584
```

Mean age of the trees using Regression: 107.4032 Standard Error: 14.31584

d. Label estimates on the graph

```
y2<-ybar
y3<-xbar
plot(x, y, type = "p", col = "blue", lwd = 2,
     xlab = "Tree Diameter", ylab = "Age of Trees", main = "Plot of Tree Age vs Size")
points(y2, y3, col = "red", pch = 16, cex = 1.0)
legend("topleft", legend = c("Data", "Ratio and Regression Means"), col = c("blue", "red"),
       pch = c(NA, 16), lwd = c(2, NA), cex = c(1, 1.5), pt.cex = 1.5)
```



Plot of Tree Age vs Size
Data
Ratio and Regression Means
Age of Trees
60
80
100
120
140
160
6
7
8
9
10
11
12
Tree Diameter

9. Using the data set agrs estimate the total number of acres devoted to farming for two domains: (a)Counties with fewer than 600 farms and (b)counties with more than 600 farms. Give standard errors for your estimates:

```
agsrs<-agsrs
head(agsrs)
```

```
##             county state acres92 acres87 acres82 farms92 farms87 farms82
## 1     COFFEE COUNTY    AL  175209  179311  194509     760     842     944
## 2    COLBERT COUNTY    AL  138135  145104  161360     488     563     686
## 3      LAMAR COUNTY    AL   56102   59861   72334     299     362     447
## 4    MARENGO COUNTY    AL  199117  220526  231207     434     471     622
## 5     MARION COUNTY    AL   89228  105586  113618     566     658     748
## 6 TUSCALOOSA COUNTY    AL   96194  120542  134616     436     521     650
##   largef92 largef87 largef82 smallf92 smallf87 smallf82
## 1       29       28       21       57       47       66
## 2       37       41       42       12       44       47
## 3        4        4        3       16       20       30
## 4       48       66       62       14       11       28
## 5        7        9        9       11       23       27
## 6       20       17       23       18       32       29
```

```
agsrssmol<-subset(agsrs %>% filter(farms92 < 600 & farms87 < 600 & farms82 < 600))
head(agsrssmol)
```

```
##             county state acres92 acres87 acres82 farms92 farms87 farms82
## 1      LAMAR COUNTY    AL   56102   59861   72334     299     362     447
## 2   COLUMBIA COUNTY    AR   57253   66305   80909     320     411     477
## 3 HOT SPRING COUNTY    AR   78498   80267   72175     419     526     512
## 4     MONROE COUNTY    AR  219444  234605  235409     278     306     369
## 5    PRAIRIE COUNTY    AR  313232  312903  301134     401     486     489
## 6     AMADOR COUNTY    CA  236222  218532  198135     367     371     359
##   largef92 largef87 largef82 smallf92 smallf87 smallf82
## 1        4        4        3       16       20       30
## 2        6        4        9       17       12       27
## 3        7        5        5       15       35       18
## 4       87       86       72        8        6        4
## 5      112      114       87        6        6        7
## 6       32       39       34       34       31       39
```

```
agsrsbig<-subset(agsrs %>% filter(farms92 > 600 & farms87 > 600 & farms82 > 600))
head(agsrsbig)
```

```
##             county state acres92 acres87 acres82 farms92 farms87 farms82
## 1     COFFEE COUNTY    AL  175209  179311  194509     760     842     944
## 2   FAULKNER COUNTY    AR  210692  223594  227593    1051    1103    1169
## 3       POLK COUNTY    AR  122871  112409  113930     791     814     840
## 4 WASHINGTON COUNTY    AR  352322  362670  345518    2539    2853    2784
## 5       NAPA COUNTY    CA  235290  258197  231037    1227    1197    1087
## 6   SISKIYOU COUNTY    CA  647446  669385  711269     689     765     796
##   largef92 largef87 largef82 smallf92 smallf87 smallf82
```

```
## 1       29        28        21        57        47        66
## 2       23        32        27        42        41        49
## 3        7         7         7        44        33        30
## 4       29        26        16       201       232       195
## 5       48        54        47       315       267       251
## 6      124       144       138        37        44        44
```

```
a1<-mean(agsrssmol$acres92)
a2<-mean(agsrssmol$acres87)
a3<-mean(agsrssmol$acres82)
xbarsmol<-(a1+a2+a3)/3
xbarsmol
```

```
## [1] 305943
```

```
n<-length(agsrssmol$county)
testsmol<-xbarsmol*n
testsmol
```

```
## [1] 43137965
```

```
c1<-mean(agsrsbig$acres92)
c2<-mean(agsrsbig$acres87)
c3<-mean(agsrsbig$acres82)
xbarbig<-(c1+c2+c3)/3
xbarbig
```

```
## [1] 319431.8
```

```
testbig<-xbarbig*n
testbig
```

```
## [1] 45039877
```

Total estimated farming land (small domain):43,137,965
Total estimated farming land (big domain): 45,039,877

```
n
```

```
## [1] 141
```

```
z<-length(agsrs$acres92)
z
```

```
## [1] 300
```

```
totsmol<-sum(agsrssmol$acres92)
SEsmol<-(z*sqrt(1-(n/z))*(testsmol/totsmol)*((25012)/sqrt(n)))
SEsmol
```

```
## [1] 465057.1
```

```
totbig<-sum(agsrsbig$acres92)
SEbig<-(z*sqrt(1-(n/z))*(testbig/totbig)*((25012)/sqrt(n)))
SEbig
```

```
## [1] 519892.5
```

Total estimated farming land (small domain):43,137,965
SE for small domain: 465057.1

Total estimated farming land (big domain): 45,039,877
SE for big domain: 519892.5

Exercises 2,4(a),5 and 8 from Chap 5

2. Senturia et.al describe a survey taken to study how many children have access to guns. . . .

(a)
Suppose the quantity of interest is percentage of the households with guns. Describe why this is a cluster sample. What is the PSU? What is the SSU? Is it a one stage or two stage? How would you estimate the percentage of households with guns, and the standard error of your estimate.

This is a cluster sample because the population of interest is the entire population of households, but of course it is not feasible to interview every household in the population to get an idea of this percentage. And an SRS is also difficult to do on such a large scale. As such this is why clustering sampling would be chosen. The above IS a cluster sample because we have clustered the population by selecting a subset of the population and extrapolating information for the whole population based on this subset. We are examining those households that are coming into the clinic, these clinics make up our primary sampling units. Since we are giving the survey to every parent regardless of the reason of their visit then this is a one stage cluster sample. To estimate the percentage of households with guns then we could just compare the number of respondents who answered yes to the ones who answered no. Alternatively we would take means of the yes and no responses for each cluster and then take the average of responses to make a ratio. Specifically we want to use the formulas for one stage clustering and use the formulas for the population.

(b)
What is the sampling population for this study? Do you think this sampling procedure results in a representative sample of households with children? Why or Why not?
The sampling population is all of the parents coming into the clinics. No, I do not think this sampling procedure is representative of households with children. At least in the US, as depending on the work situation of the parents, the children may or may not have health care. In this case all children who live in households with guns but don't receive healthcare due to cost constraints will be under represented in this sample. Likewise, children who live with parents who addicted to drugs or alcohol and spend most of the day inebriated will be unlikely to receive regular checkups. And then of course parents who are not around, due to work such as on a graveyard shift, or are just inattentive will be unlikely to be represented. However in all of these cases, per personal experience, just because someone is in one of the above situations does not mean that they do not spend an obscene amount of money on firearms. As such, this sample will probably misrepresent the actual percentage of households with children that have guns, as it is only going to be looking at households that bother to bring their kids in to the clinic.

4.

Survey evidence is often introduced in court cases involving trademark violation and employment discrimination…

(a)Explain why this is a cluster sample.

This is a cluster sample because instead of taking an SRS of all magazines of interest we are taking a specific year, in this case 1988, and then taking an SRS from the publications released in that year. Additionally, if ignoring the year then the 25 magazines serves as the cluster or PSU and all of the articles count as the SSU.

5.
A language school owner takes an SRS of 10 of the 72 introductory Spanish courses. . . .
(a)
Estimate the total number of students planning a trip to the Spanish speaking country in the next year and give a 95% CI

```r
spanish <- read_csv("C:/Users/super/Downloads/spanish.csv")
```

```
## Rows: 196 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (3): class, score, trip
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(spanish)
```

```
## # A tibble: 6 x 3
##   class score  trip
##   <dbl> <dbl> <dbl>
## 1    34    57     0
## 2    34    69     0
## 3    34    61     0
## 4    34    62     0
## 5    34    42     0
## 6    34    45     0
```

```r
takingtrip<-spanish$trip[spanish$trip==1]
length(takingtrip)
```

```
## [1] 63
```

```r
196/10
```

```
## [1] 19.6
```

```r
19.6*72
```

```
## [1] 1411.2
```

```r
63/196
```

```
## [1] 0.3214286
```

```r
0.32*1411.2
```

```
## [1] 451.584
```

The number of students from the sample planning to take a trip are 63.
The estimated total number of students planning to take a trip is 452 students.
The 95% CI is (403.4,501.6)
(b)
Estimate the mean vocabulary test score for Introductory Spanish students in the language school and give a 95%

```
mean(spanish$score)
```

```
## [1] 66.79592
```

```
66.79592
```

```
## [1] 66.79592
```

```
Mi<-spanish$class
Mi
```

```
##   [1] 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 65 65 65
##  [26] 65 65 65 65 65 65 65 65 65 65 65 65 65 60 60 60 60 60 60 60 60 60 60 60 60
##  [51] 60 60 60 60 60 60 60 60 60 60 60 60 60 60 69 69 69 69 69 69 69 69 69 69 69
##  [76] 69 69 69 69 69 69 69 69 69 69 69 68 68 68 68 68 68 68 68 68 68 68 68 68 68
## [101] 68 68 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 47 47
## [126] 47 47 47 47 47 47 47 47 47 47 47 47 47 47 47 47 23 23 23 23 23 23 23 23 23
## [151] 23 23 23 23 23  4  4  4  4  4  4  4  4  4  4  4  4  4  4  4  4  4  4  4  4
## [176] 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39 39
```

```
ybar<-c()
for(i in 1:length(Mi)){ybar[i]<-66.79592*Mi[i]}
sum(ybar)/sum(Mi)
```

```
## [1] 66.79592
```

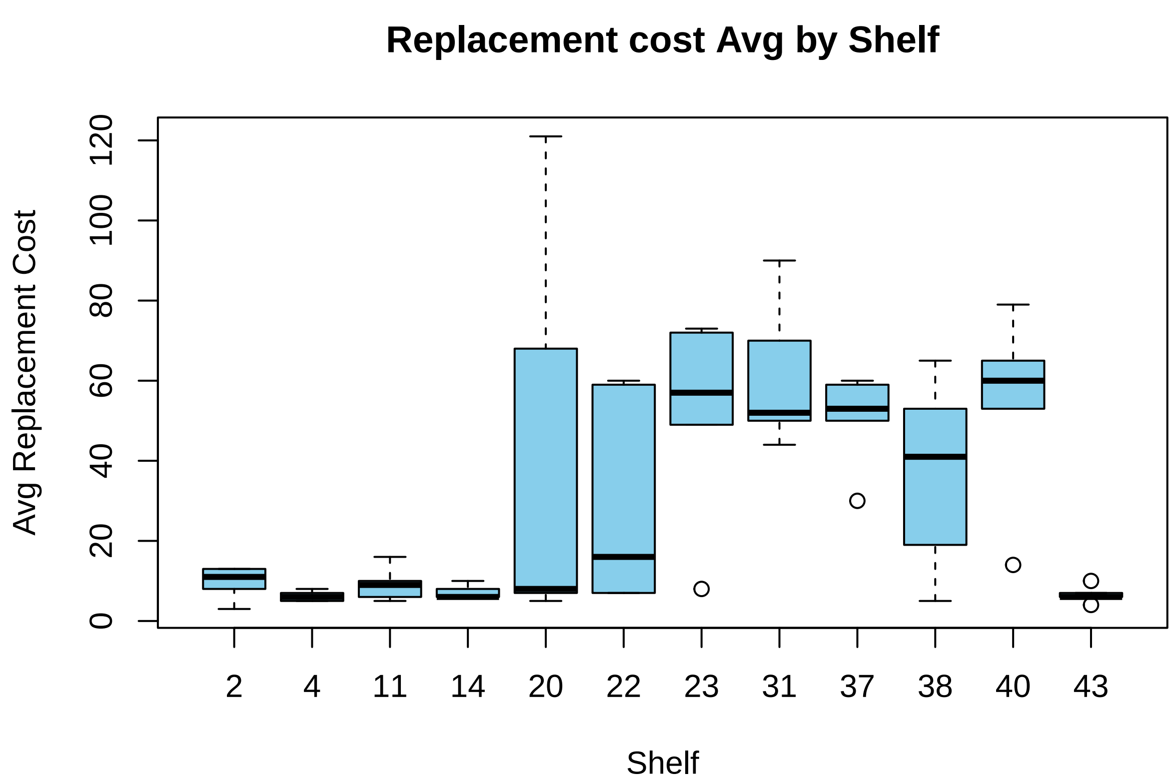
The mean vocabulary test score for Introductory Spanish students is 66.8.
The 95% CI is (63.4,70.5)

8.A homeowner with a large library needs to estimate the purchase cost and replacement value of the book collection for insurance purposes. She has 44 shelves containing books and selects 12 at random. . . .
(a)
Draw side-by-side boxplots for the replacement cost of the books on each shelf. Does it appear that the means are about the same? The variances?

```
books<-books
means_replace <- aggregate(replace ~ shelf, data = books, FUN = mean)
means_replace
```

```
##    shelf replace
## 1      2     9.6
## 2      4     6.2
## 3     11     9.2
## 4     14     7.2
## 5     20    41.8
## 6     22    29.8
## 7     23    51.8
## 8     31    61.2
## 9     37    50.4
## 10    38    36.6
## 11    40    54.2
## 12    43     6.6
```

```
boxplot(books$replace~books$shelf, col="skyblue", xlab="Shelf", ylab="Avg Replacement Cost", main="Repla
```



The means and the variances for each shelf seem to differ by a large amount. Some shelves average around

10, while others average 50, some have small or no tails, some have tails to go to the maximum value while the mean is close the minimum. Overall I would say that no, neither the means nor the variances appear to be the same from shelf to shelf.

(b)
Estimate the total replacement cost of the library and find the standard error of your estimate. What is the estimated coefficient of variation?

```
cost1<-sum(books$replace)
t_hat<-c()
for(i in 1:length(means_replace)){t_hat[i]<-means_replace[i]*5}
t_hat
```

```
## [[1]]
##  [1]  10  20  55  70 100 110 115 155 185 190 200 215
##
## [[2]]
##  [1]  48  31  46  36 209 149 259 306 252 183 271  33
```

```
t_hat1<-c(48,31,46,36,209,149,259,306,252,183,271,33)
M_i<-5*12
y_bar_hat<-(sum(t_hat1))/(sum(M_i))
y_bar_hat
```

```
## [1] 30.38333
```

```
ti.minus.Mi<-c()
for(i in 1:length(t_hat1)){ti.minus.Mi[i]<-(t_hat1[i]-(5*y_bar_hat))^2}
ti.minus.Mi
```

```
##  [1] 10798.673611 14620.840278 11218.340278 13436.673611  3258.506944
##  [6]     8.506944 11466.840278 23741.673611 10016.673611   966.173611
## [11] 14180.840278 14141.173611
```

```
s_sqr<-(1/(60-1))*sum(ti.minus.Mi)
44*60*y_bar_hat
```

```
## [1] 80212
```

The total estimated replacement cost of the library is $80212, with an average cost of $30.38 per book. The standard error is 8.87313.

(c)
Estimate the average replacement cost per book along with the standard error.
The average replacement cost per book as found in the last step is $30.38 per book.